

Project Proposal

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Bluetooth-Controlled Music Player

For our final project, we plan to build a Bluetooth-controlled music player using the TM4C123 LaunchPad. The board will store around ten short songs, and each song will be played on the buzzer by generating square waves at the correct note frequencies, which builds on the timer-based sound generation concepts we learned in Lab 2. To select a song, we will send a single character command ('1' through '9' and '0') from a phone over Bluetooth. For the wireless link we chose the HM-10 module instead of the HC-05 from the project ideas list, because the HC-05 uses Bluetooth Classic which iPhones do not support, while the HM-10 uses Bluetooth Low Energy and works with iOS. The HM-10 will communicate with the Tiva over UART, applying the serial communication concepts from Lab 6, along with receive interrupts and the NVIC, similar to what we covered in Lab 3. The buzzer output relates back to the GPIO fundamentals from Lab 1, and if time and available pins permit, we may also add LEDs through the GPIO ports that light up along with the notes as a simple visualizer.